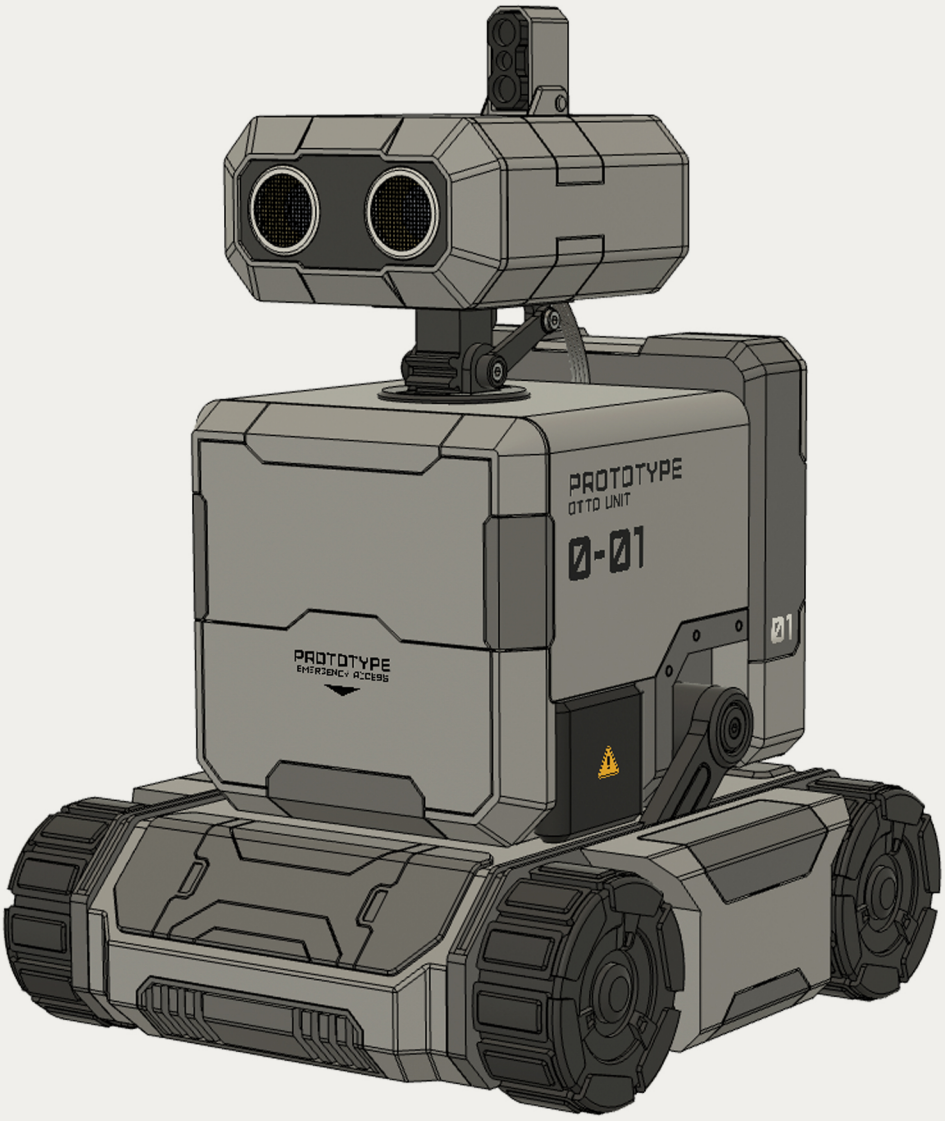
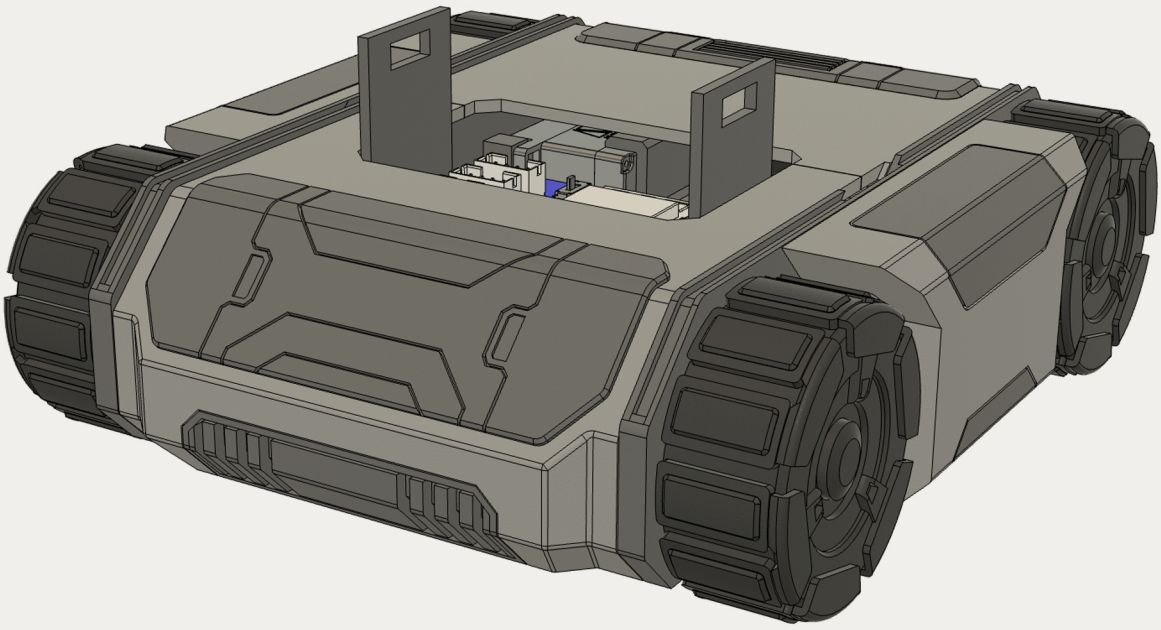
OTTO UNIT 01

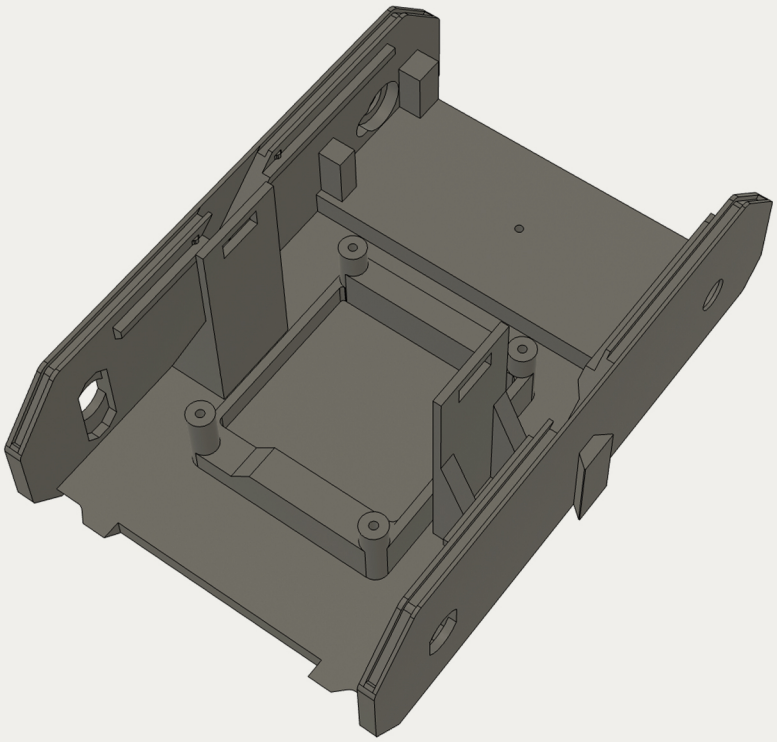


ASSEMBLY INSTRUCTIONS

PART 1: WHEEL BASE



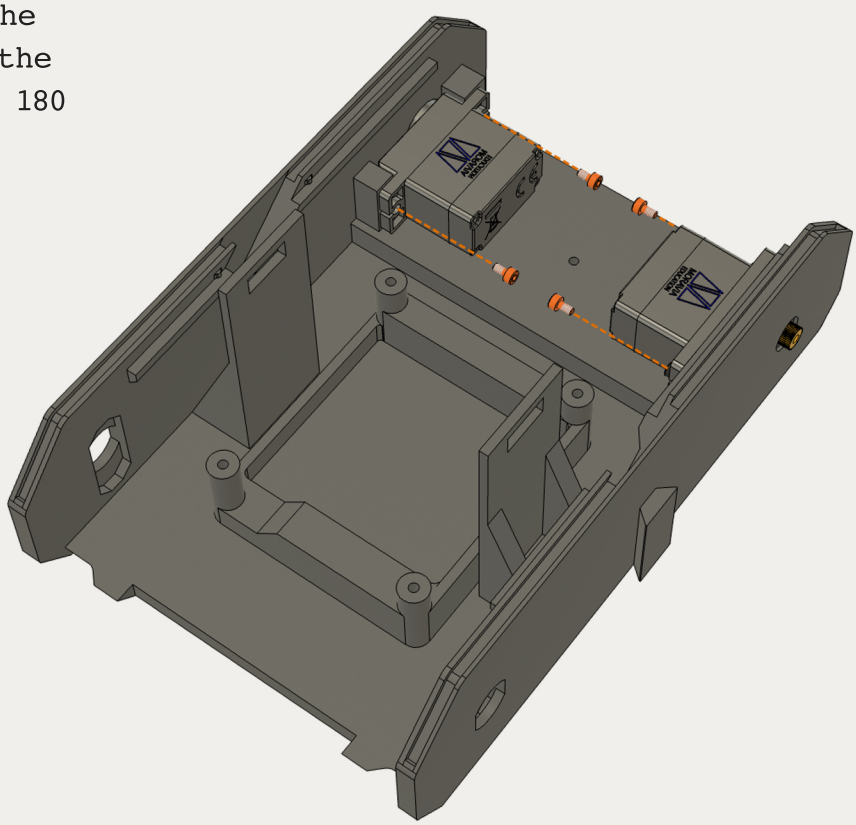
1 CHASSIS



2 WHEEL SERVOS

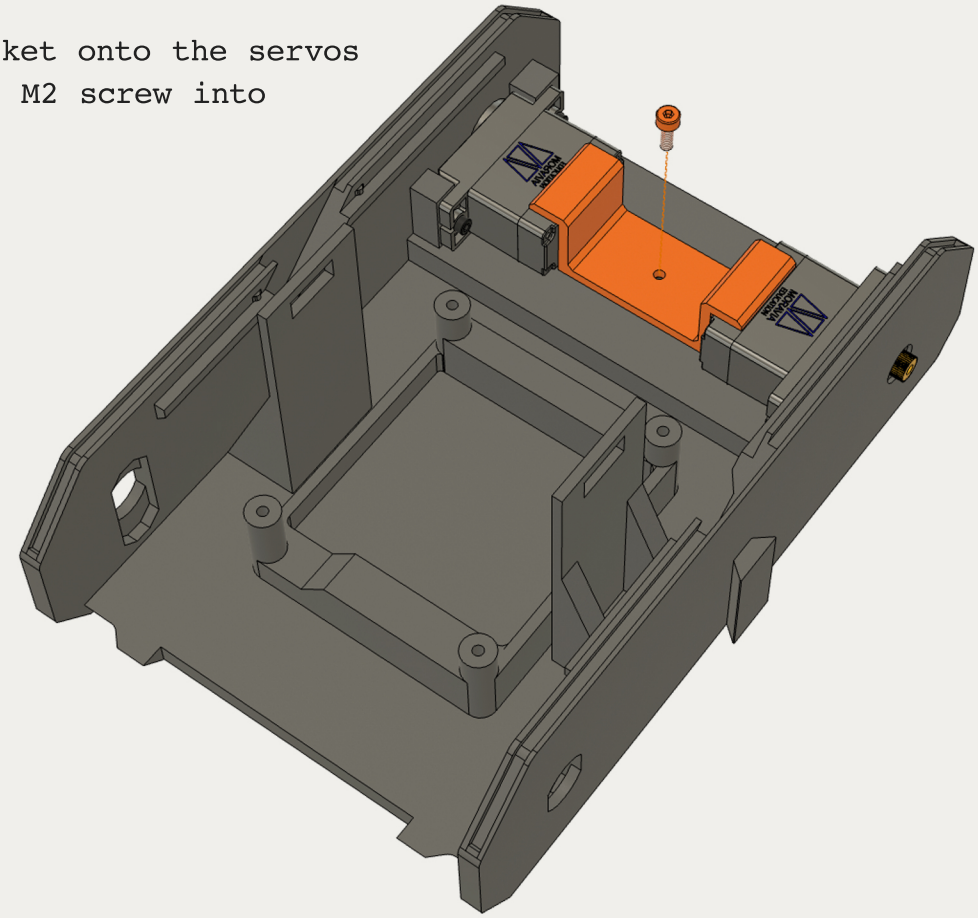
Insert the motor servos into the holes at the back of the chassis. Make sure to use the 360 degree servos, not the 180 angle servos.

Use the screws that came with your servos to hold them in place.



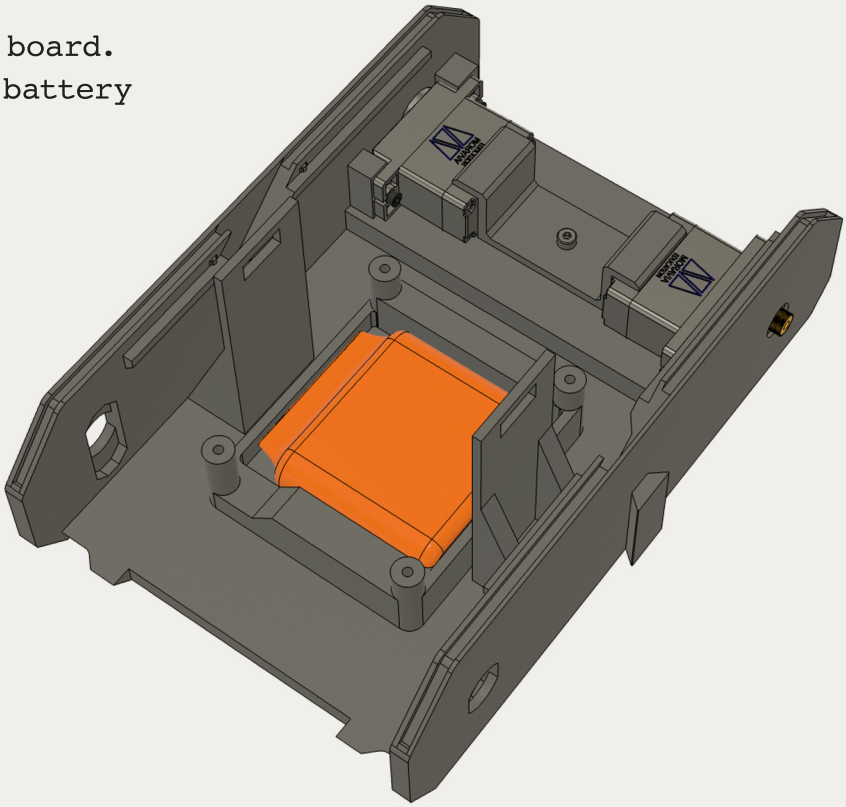
3 SERVO BRACKET

Place the servo bracket onto the servos and tightly screw an M2 screw into the hole.



4 BATTERY

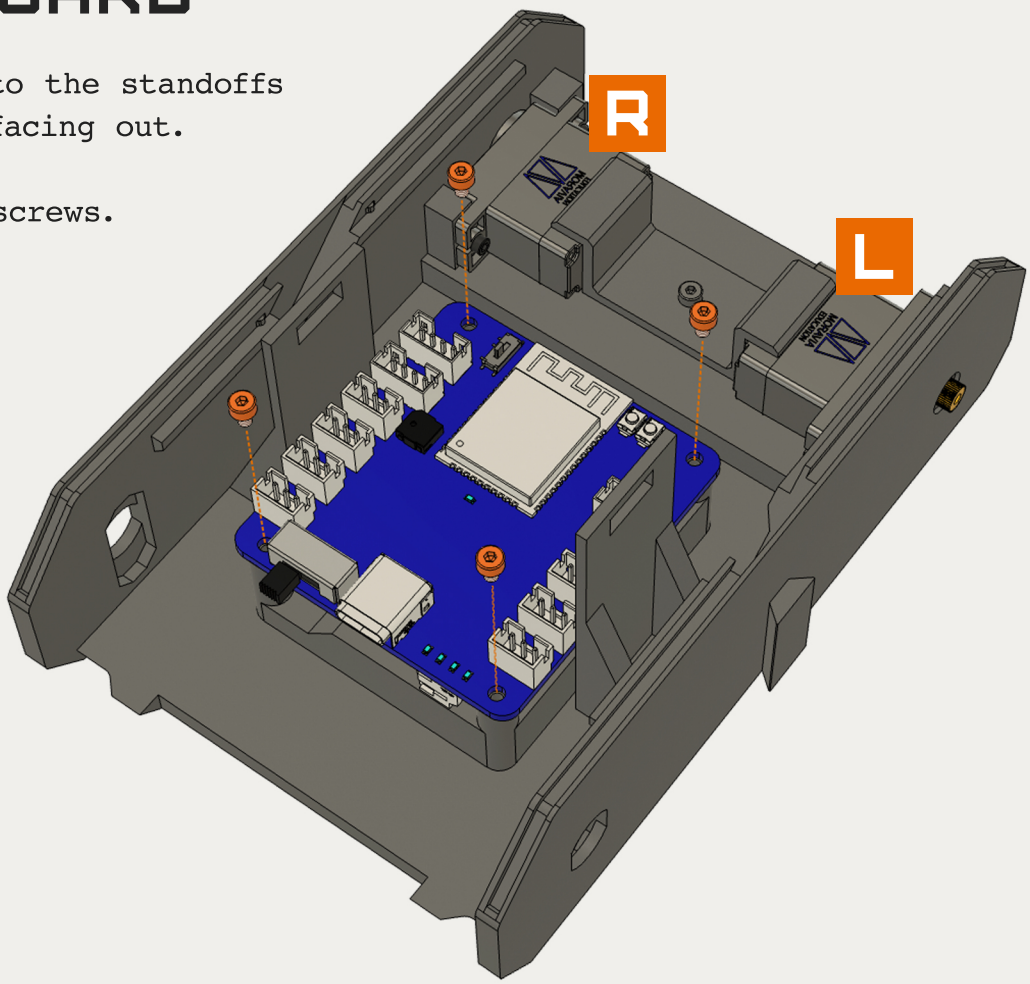
Connect battery to the main board. Then place the battery into battery holder.



5 MAIN BOARD

Place main board onto the standoffs with usb connector facing out.

Screw down with M2 screws.

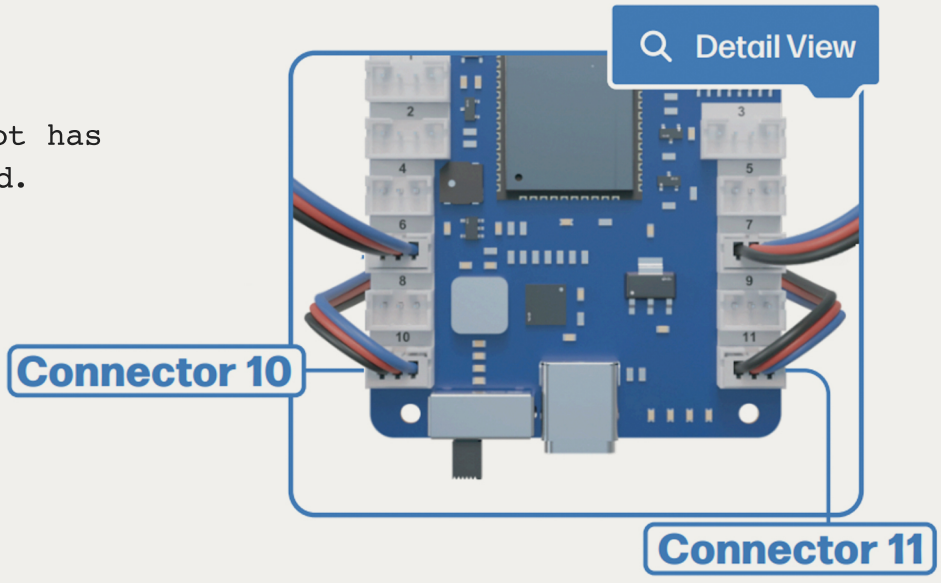


Next, plug motors to connectors 10 (L) and 11 (R).

CAUTION

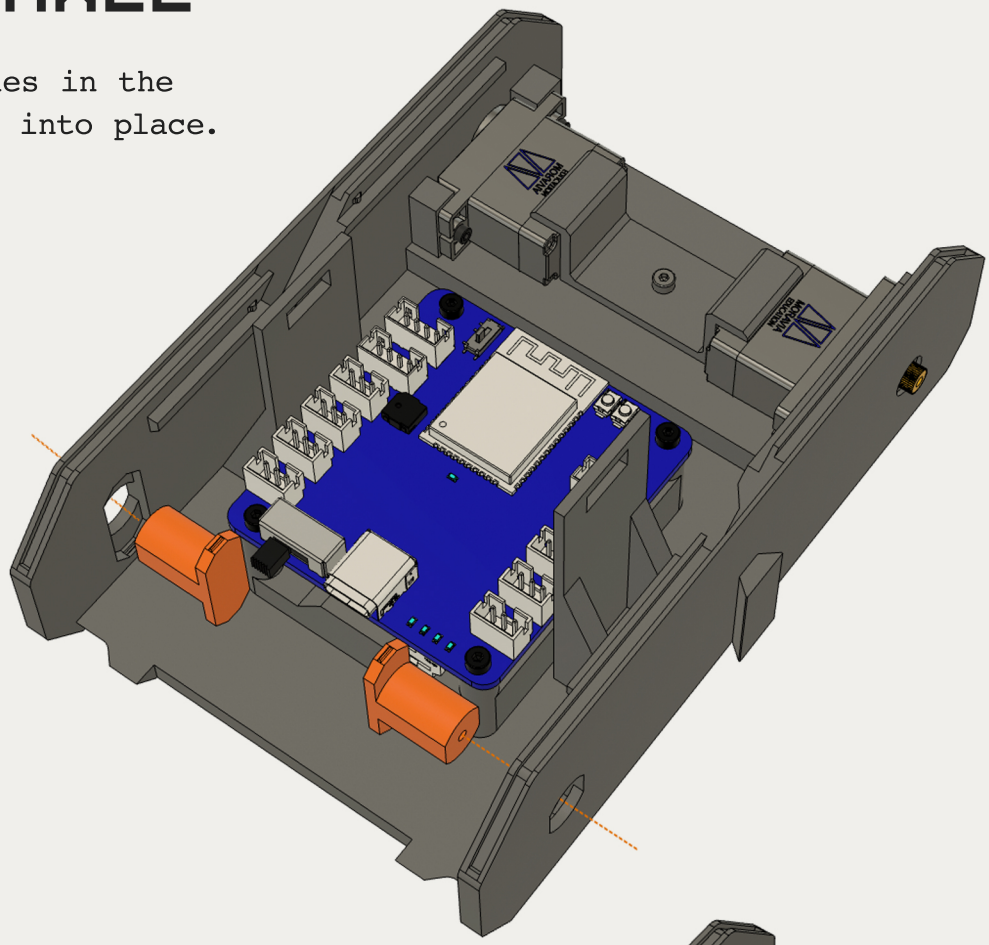
The original otto robot has the main board rotated.

Make sure to connect:
Left motor to 10
Right motor to 11



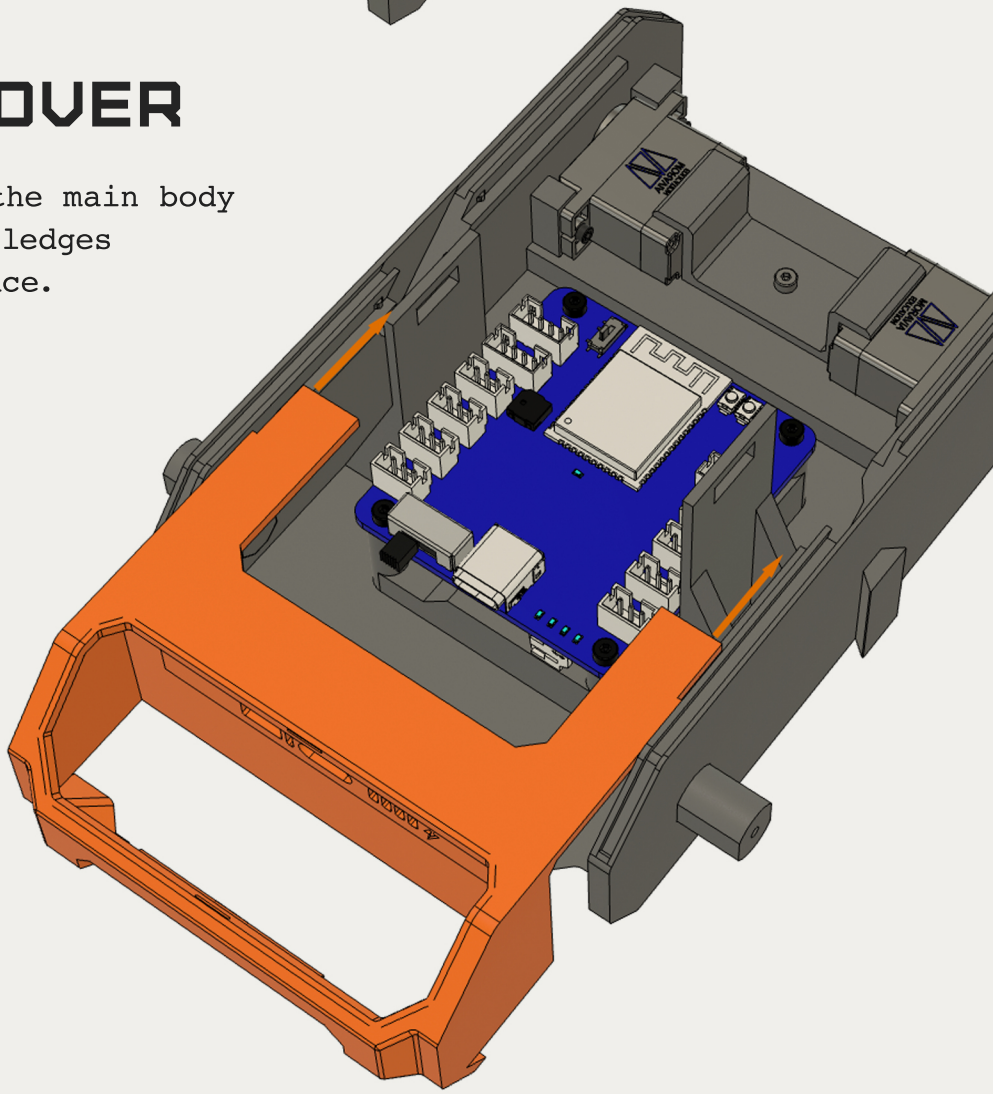
6 WHEEL AXLE

Slide front wheel axles in the holes until it clicks into place.



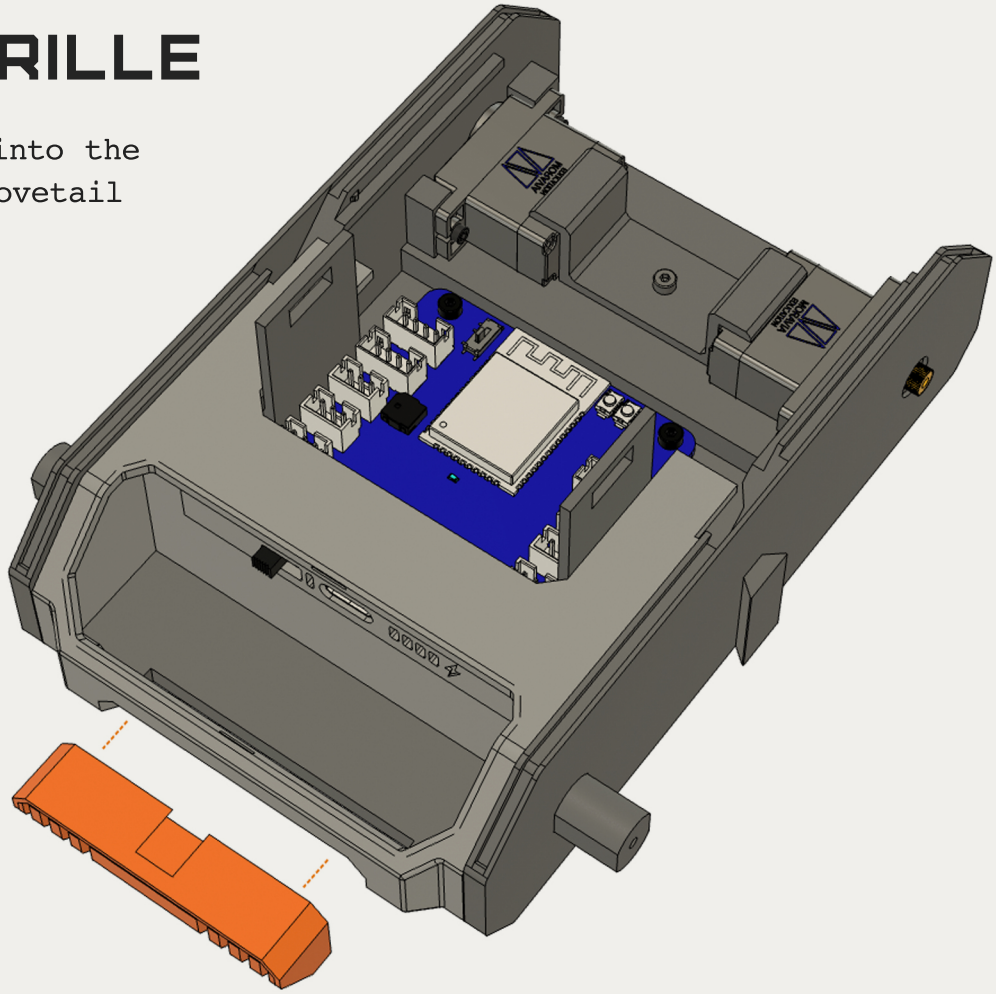
7 FRONT COVER

Slide front cover into the main body where it sits on little ledges until it clicks into place.



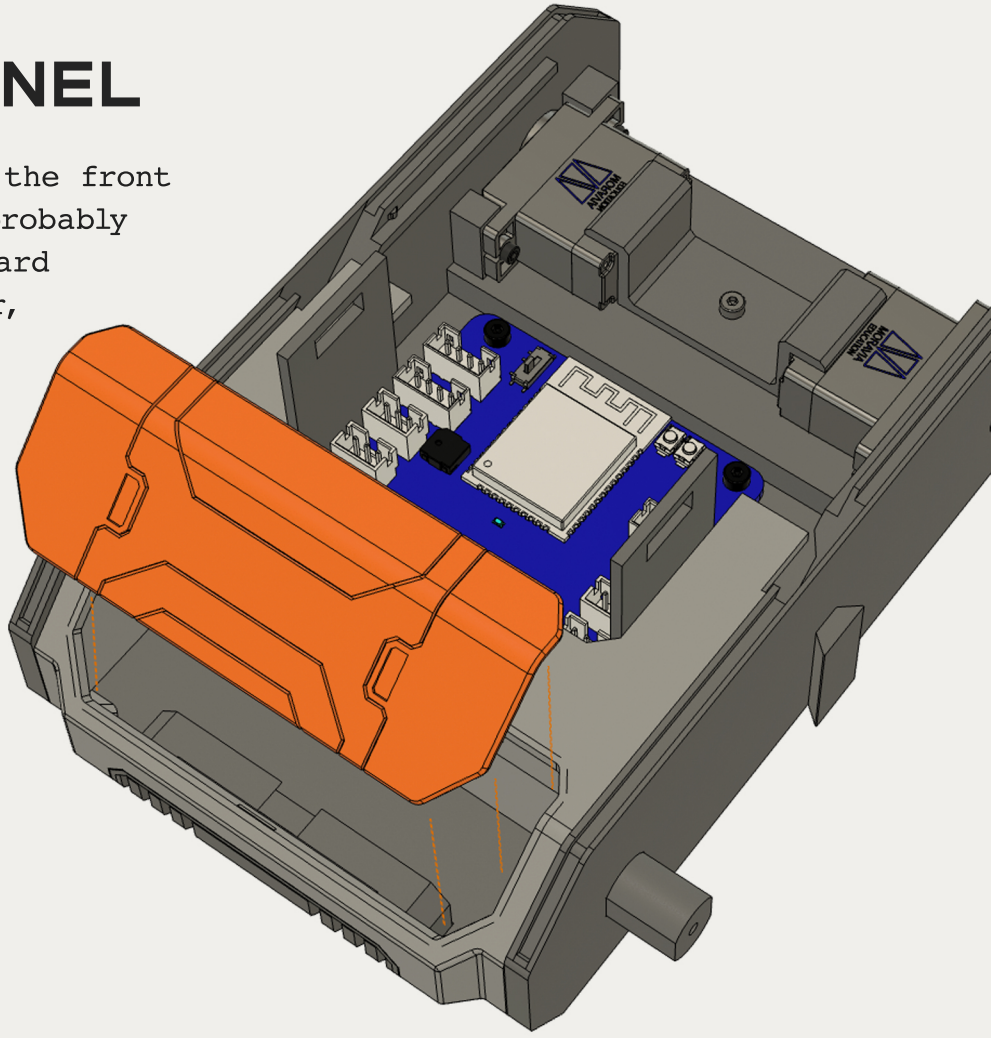
8 FRONT GRILLE

Slide the front grille into the front cover using the dovetail connector.



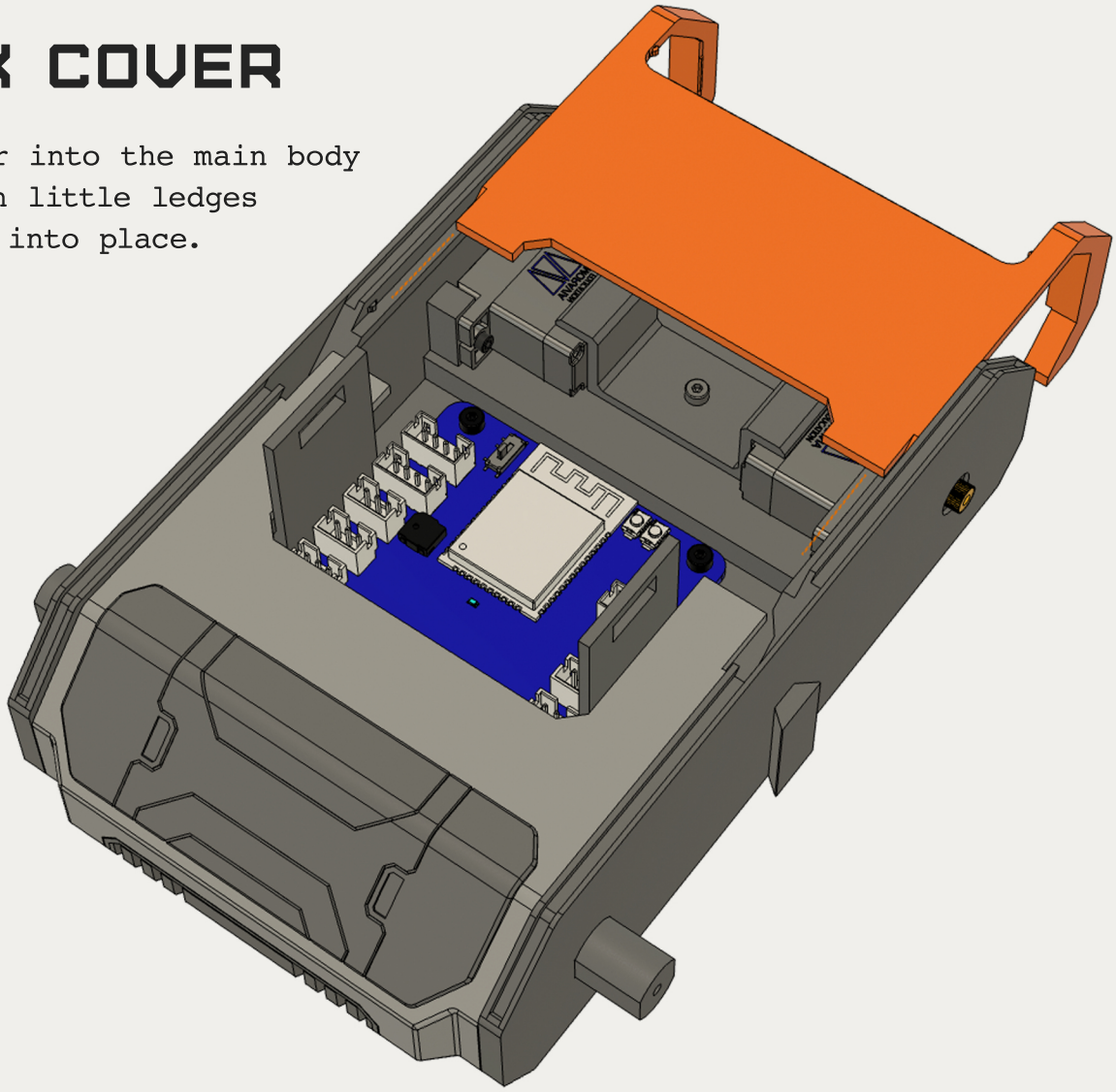
9 FRONT PANEL

OPTIONAL: You can clip in the front panel, however, you will probably need to access to main board switch and usb c connector, so you can skip this step until later.



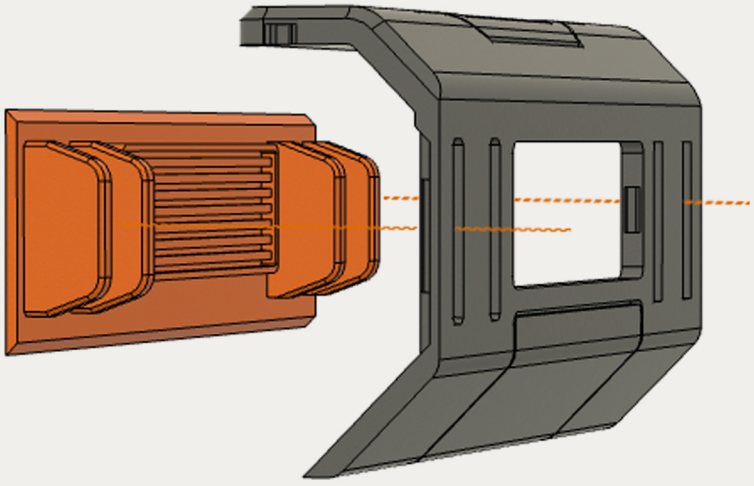
10 BACK COVER

Slide back cover into the main body where it sits on little ledges until it clicks into place.



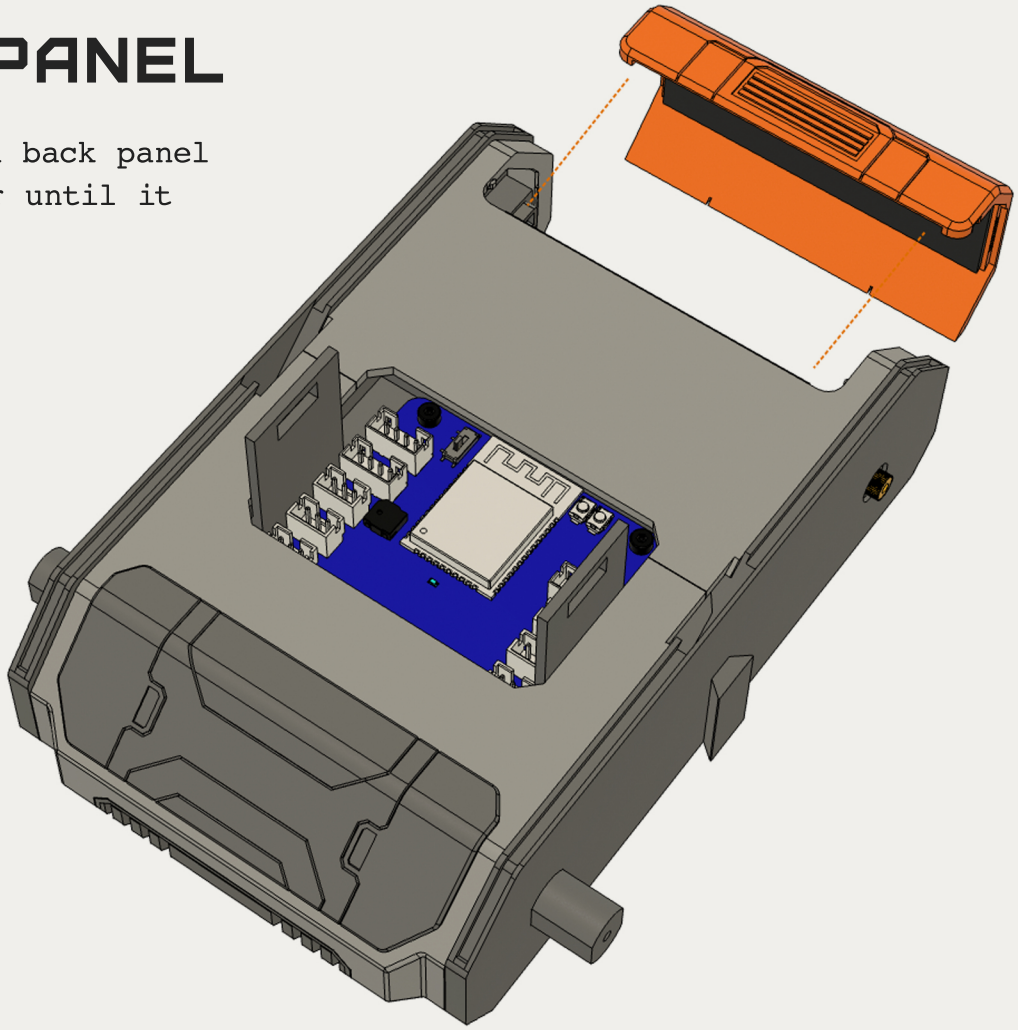
11 BACK PANEL ASSEMBLY

Slide the decorative panels together until it clicks into place.



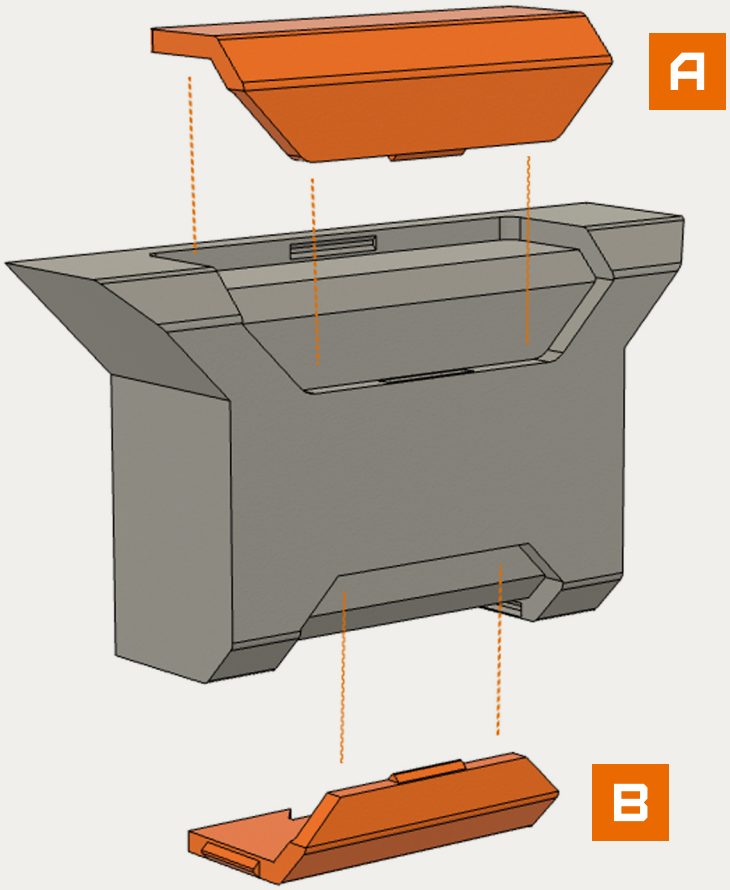
12 BACK PANEL

Slide the assembled back panel into the back cover until it clicks into place.



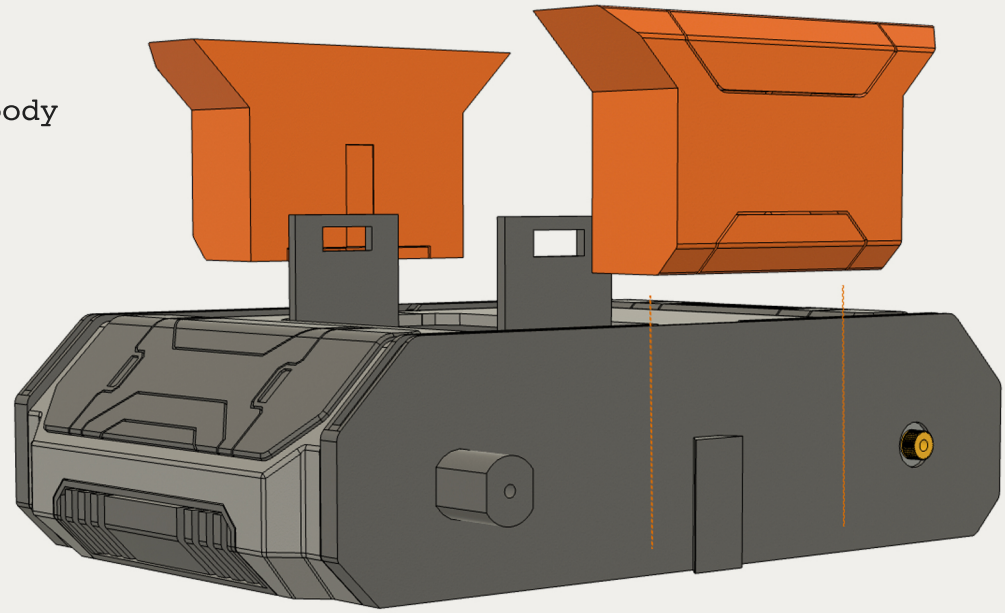
13 SIDE PANELS

Click A and B panels into the side part.



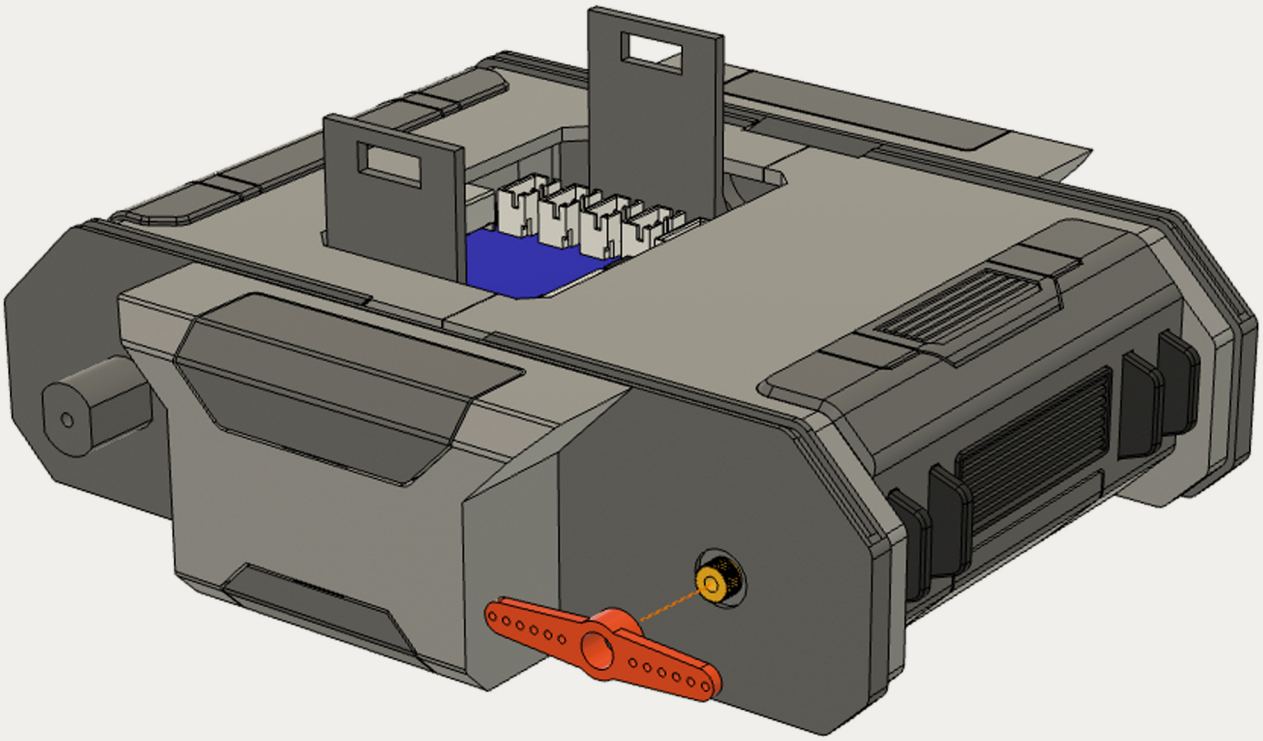
14 SIDE PARTS

Slide the side parts down onto the main body using the dovetail connector.



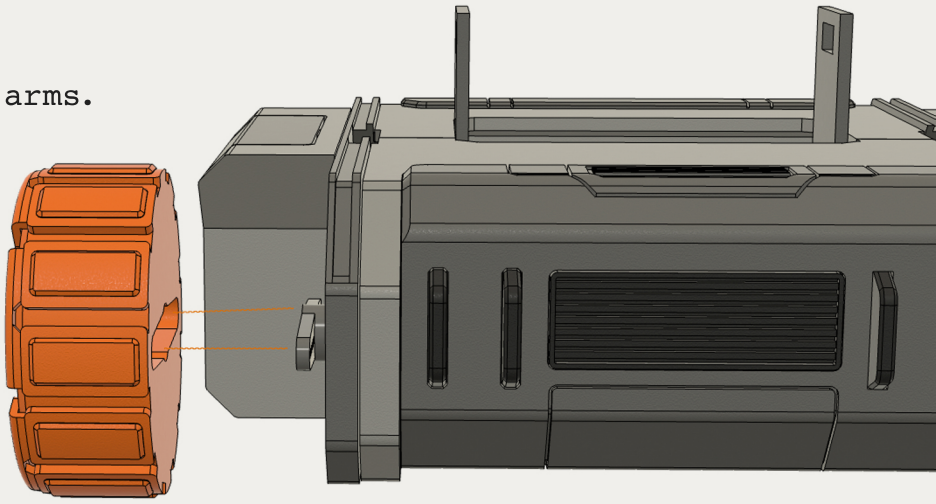
15 SERVO ARMS

Push the motor arms onto the servo axles.

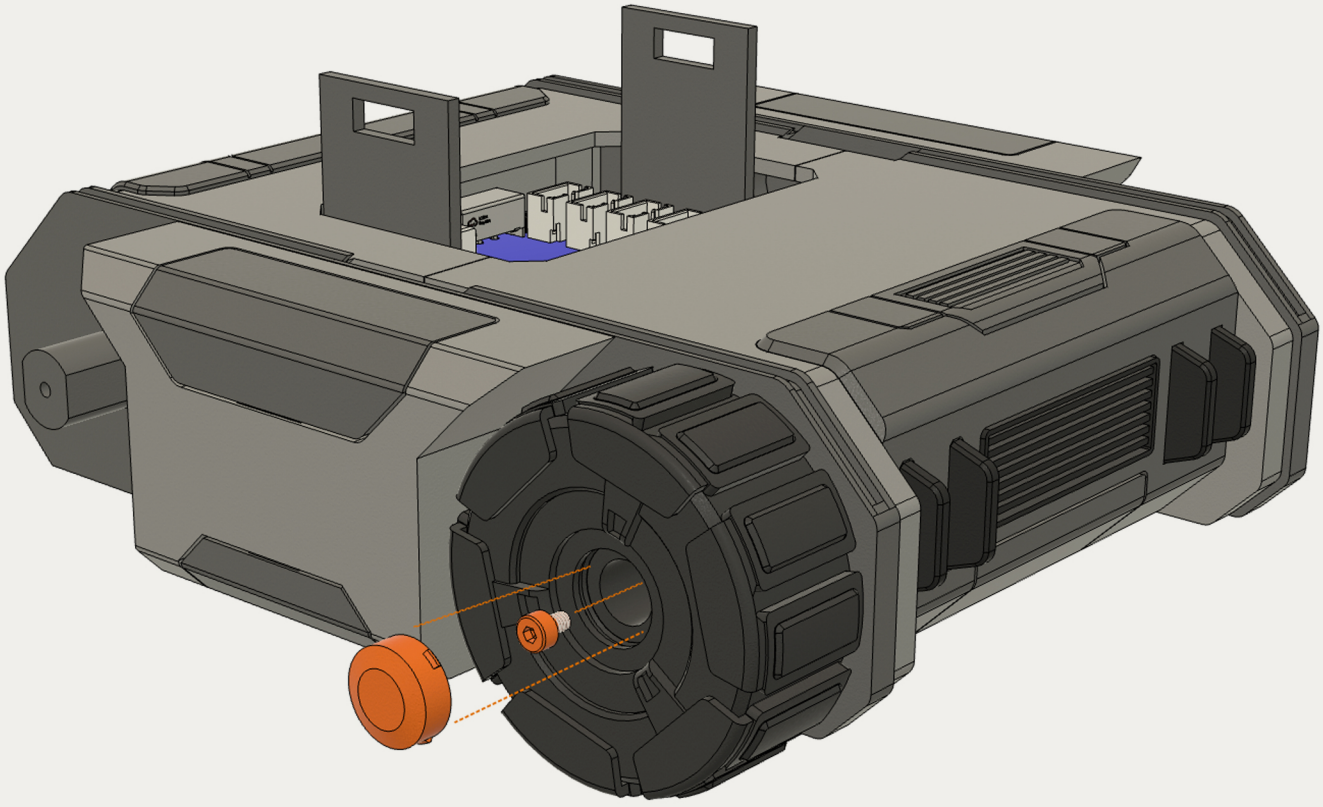


16 WHEEL ASSEMBLY

Slide the back wheels onto
the previously mounted servo arms.

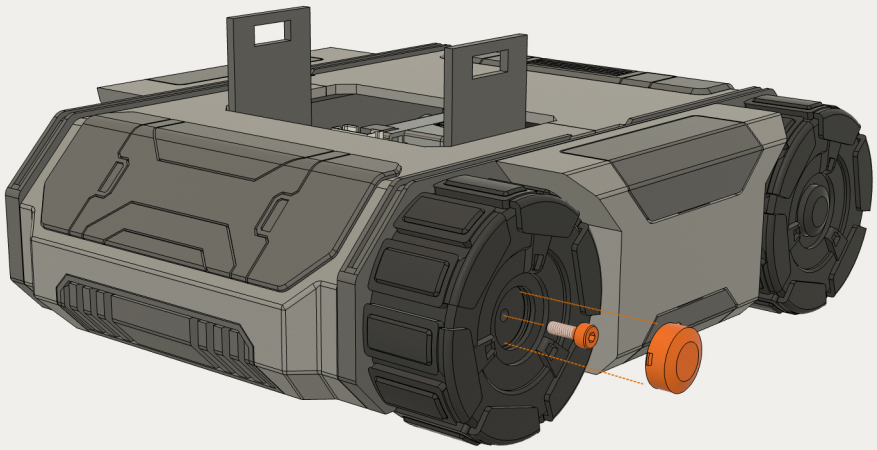


Screw in the self tapping screw that came with your servos and
then cover the hole with a cover.

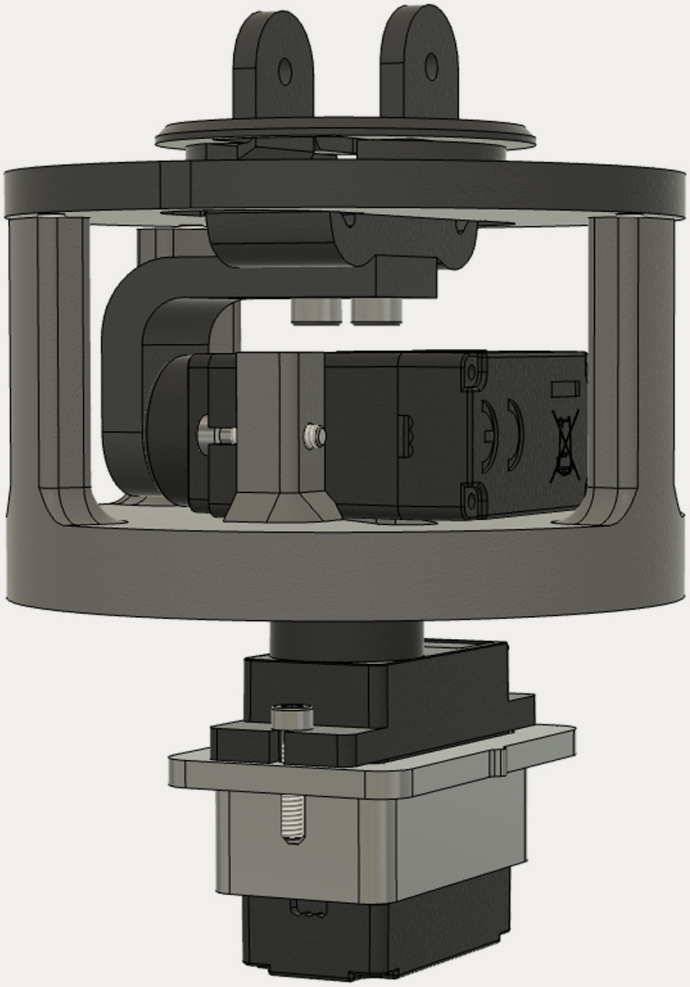


17 FRONT WHEELS

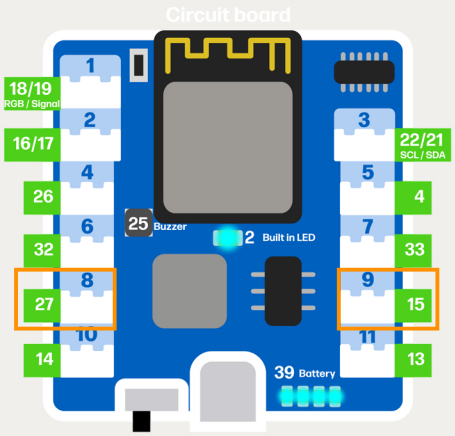
Screw M2 screws into the front wheels.
Do not overtighten - the wheels must move freely.
Lastly, place the cover over the screw.



PART 2: TILT-PAN



1 CENTERING SERVOS



Before you start, connect you 180 angle servos into connectors 8 and 9.

Next we want to “center” these servos to a 90 degree angle. Use either the HP block-based web app or the provided micropython code.

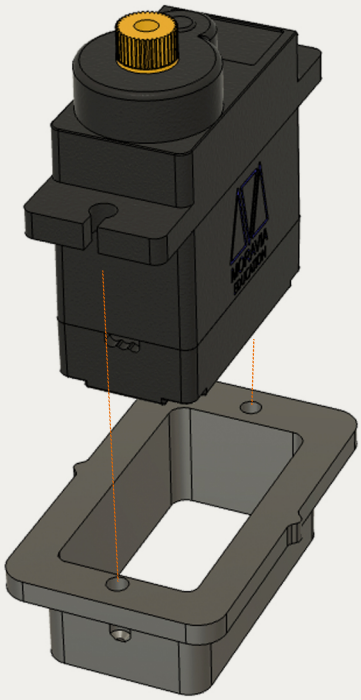
The robot doesn’t use the full +-90 degree rotation, so don’t worry if you move it a bit during assembly - the center position can then be offset in code.

When done, unplug the servos and continue with the tilt-pan mechanism assembly.

2 PAN SERVO

The pan servo will move the head side to side. Before mounting mark the end of the cable “PAN”.

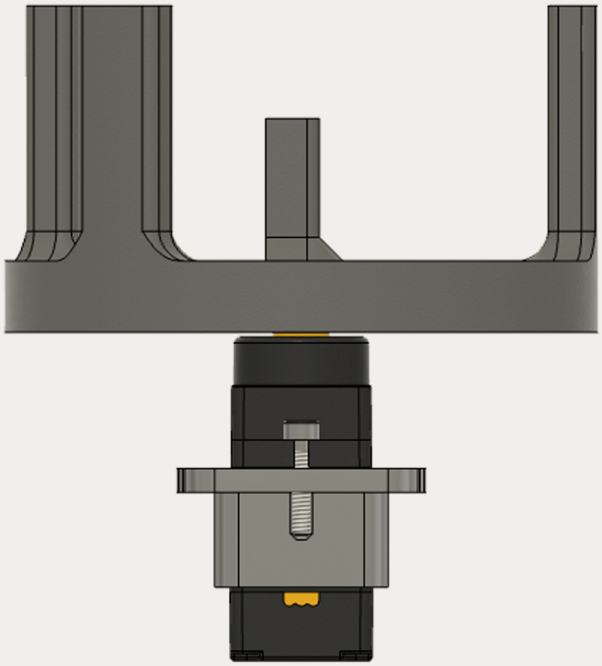
Slide the servo into the mounting bracket and tighten with provided self tapping screws.



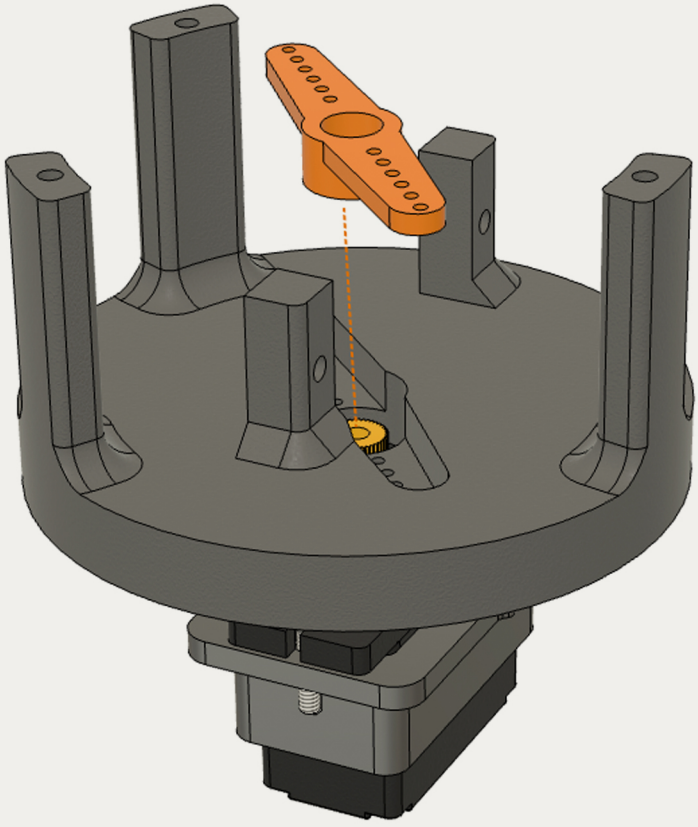
FRONT

3 ROTATION BRACKET

Place the rotation bracket on the servo, and try to align the little servo brackets with the servo.



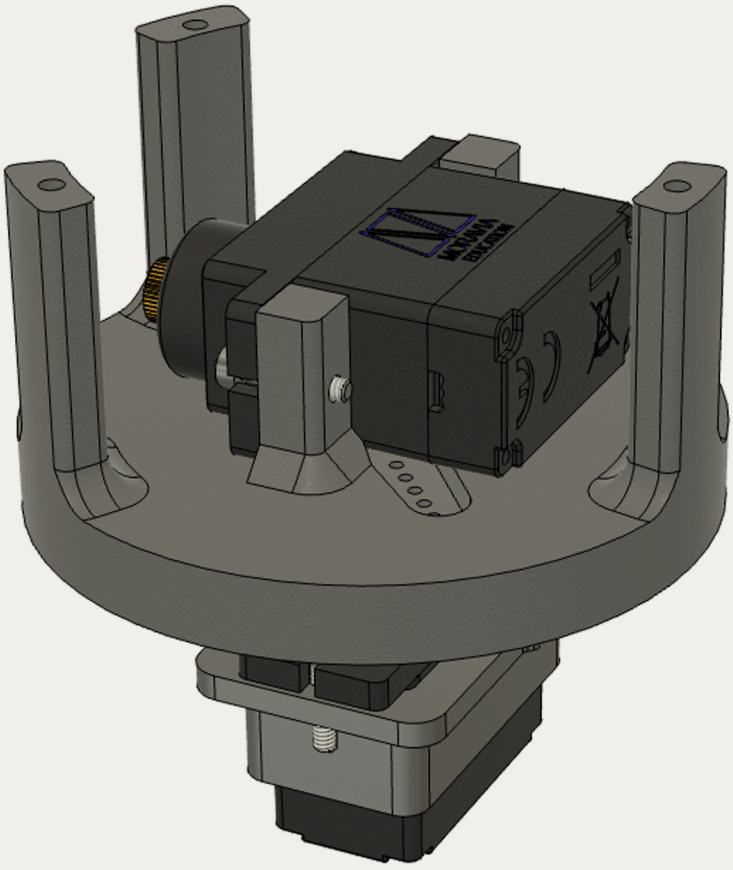
Push the servo arm onto the servo axle and tighten with provided self tapping screw.



4 TILT SERVO

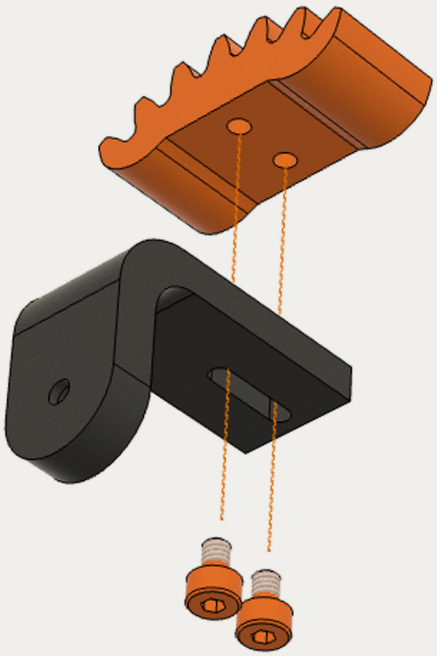
The tilt servo will move the head up and down. Before mounting mark the end of the cable “TILT”.

Slide the servo into the mounting bracket and tighten with provided self tapping screws.



5 TILT GEAR ARM

Use two M2 screws to assemble the tilt gear arm. The slot allows for slight adjustments, but for now, try to align the gear to the edge of the arm.

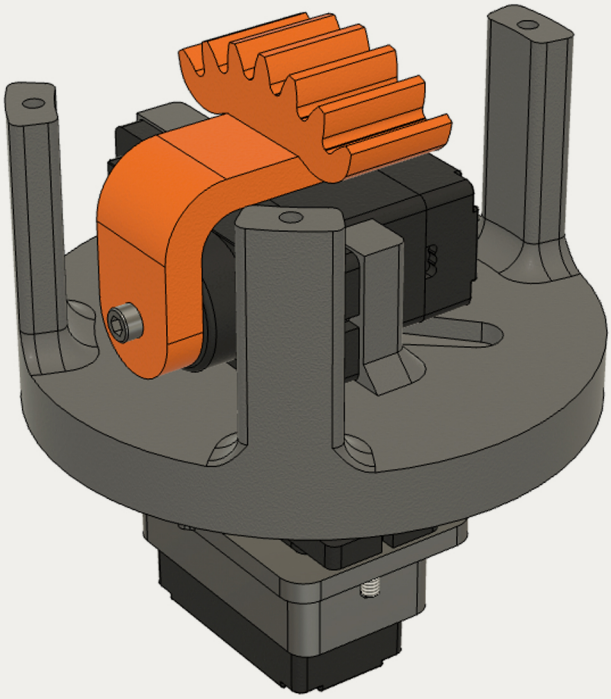


NOTE

There are several .stl files of the arm with different sized openings. Measure your servo axle and pick the matching one.

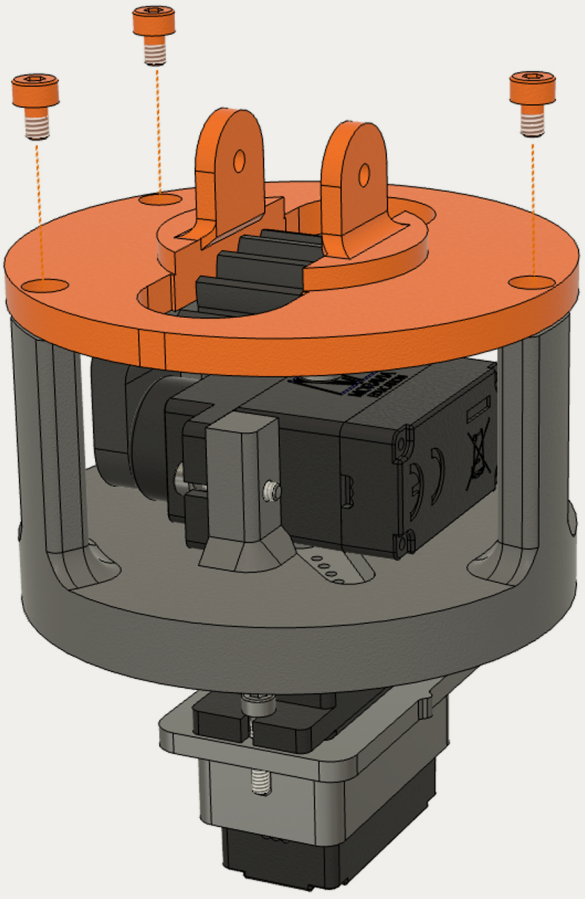
6 CONNECT GEAR ARM

Place gear arm on the tilt servo and tighten with provided self-tapping screw.



7 MOUNT TOP COVER

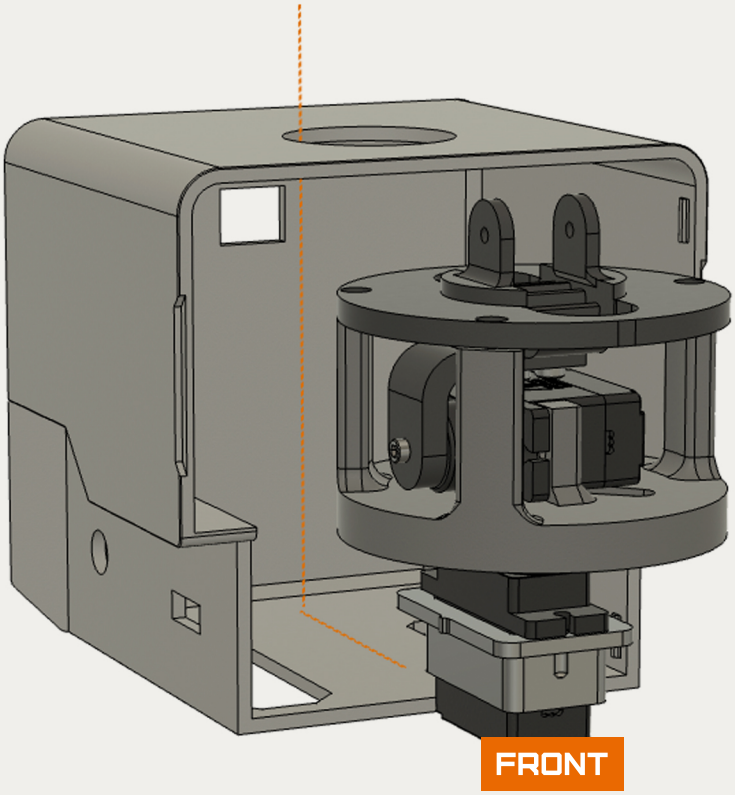
Place top cover on the standoffs and screw using M2 screws.

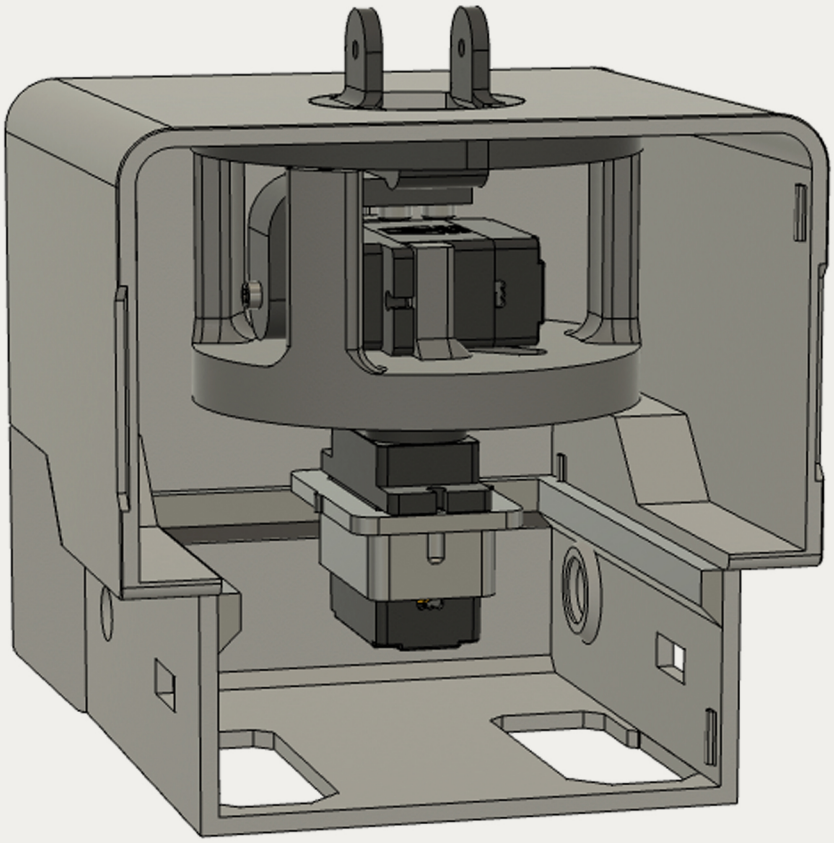


PART 3: MAIN BODY

1 INSERT TILT-PAN MECHANISM

Slide the tilt-pan mechanism IN and UP.

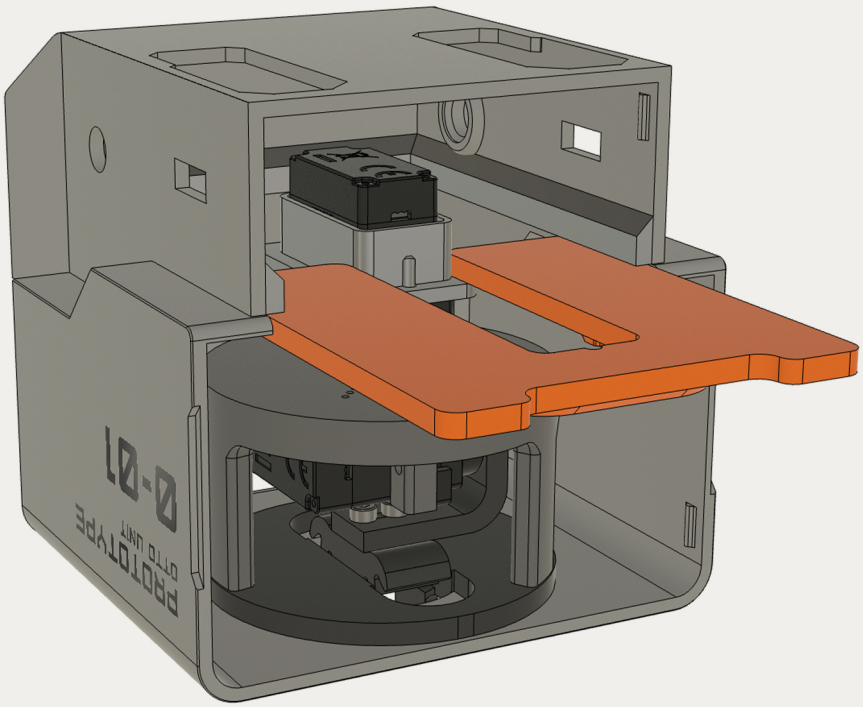




2 CLIP IN MECHANISM

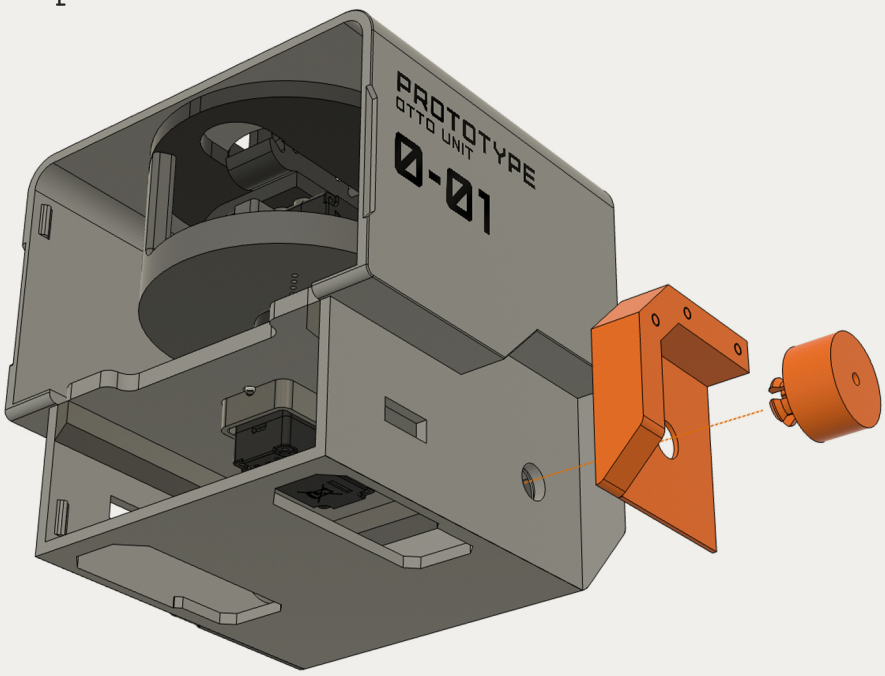
Carefully rotate the whole body upside down.
Slide the mounting plate between the body's shelf
and the servo mounting bracket.

Put a finger behind the servo and click the plate in.

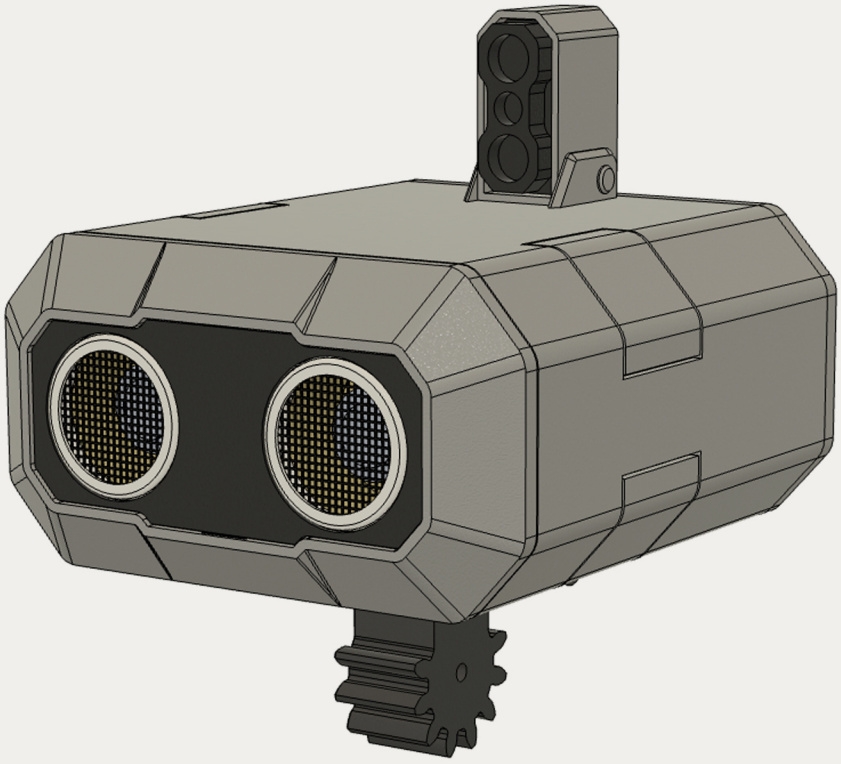


3 SIDE ARM MOUNT

Rotate the body back up. On both sides, place the cover part onto the body and clip in the side arm mount which will hold the cover in place.

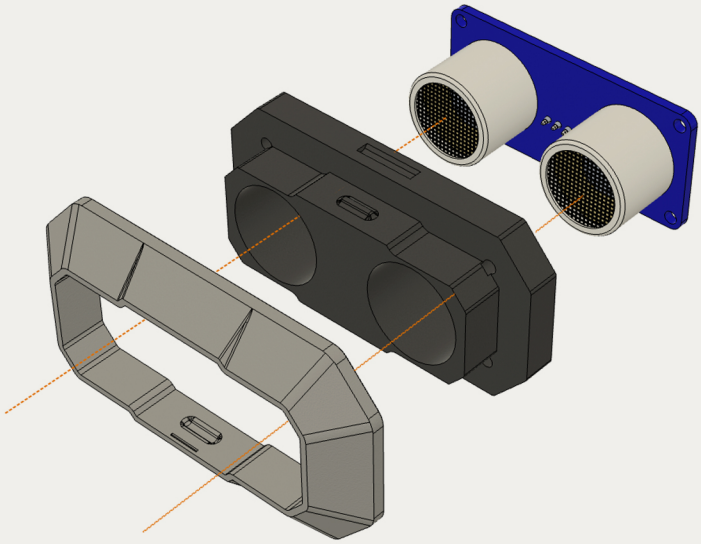


PART 4: HEAD



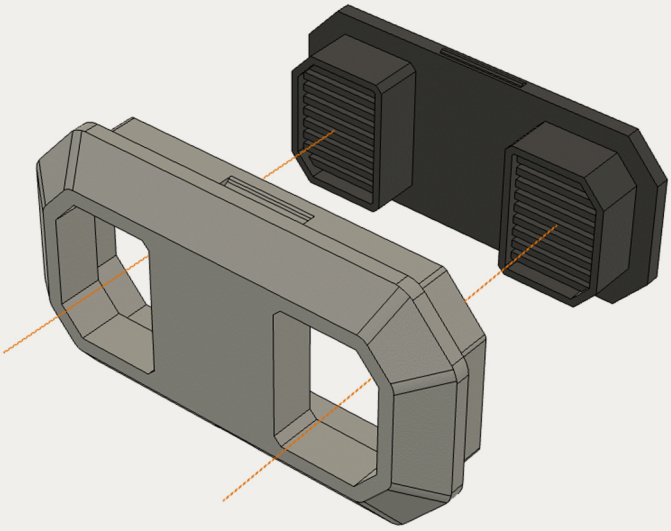
1 ASSEMBLE FRONT

Slide the eye piece to the front plate until it clicks. Then slide the ultrasonic sensor in and hold in place with self tapping screws.



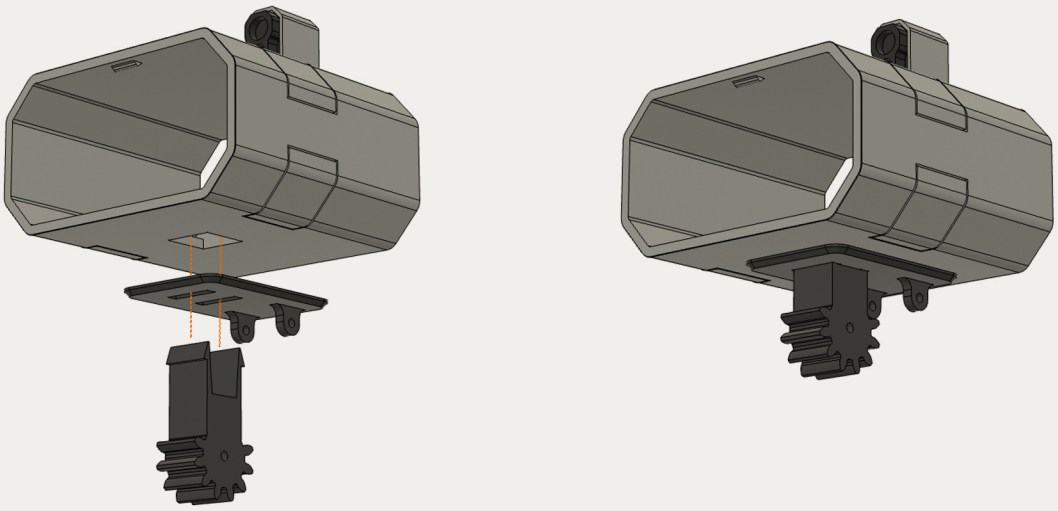
2 ASSEMBLE BACK

Slide decorative vents into the back part until it clicks into place.



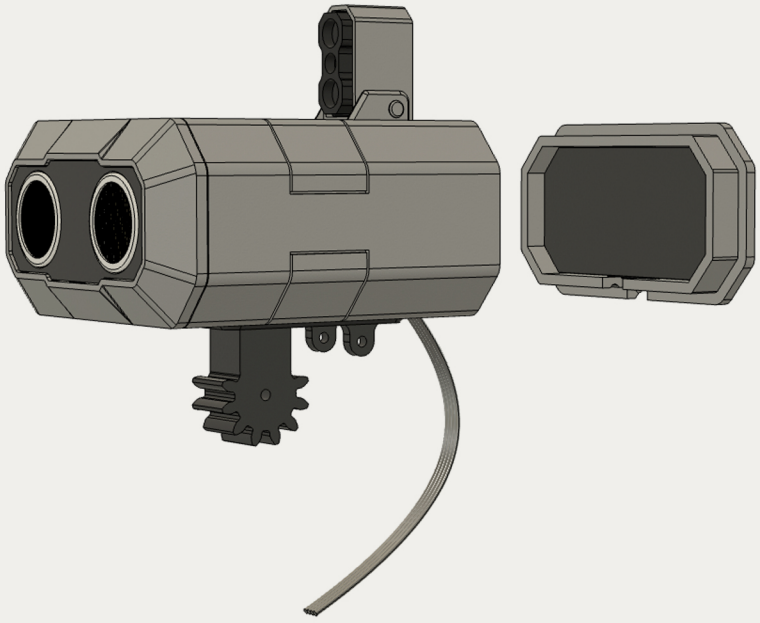
3 ASSEMBLE MIDDLE

Take the gear and slide the clips into the plate.
Then slide the whole thing into the middle part.

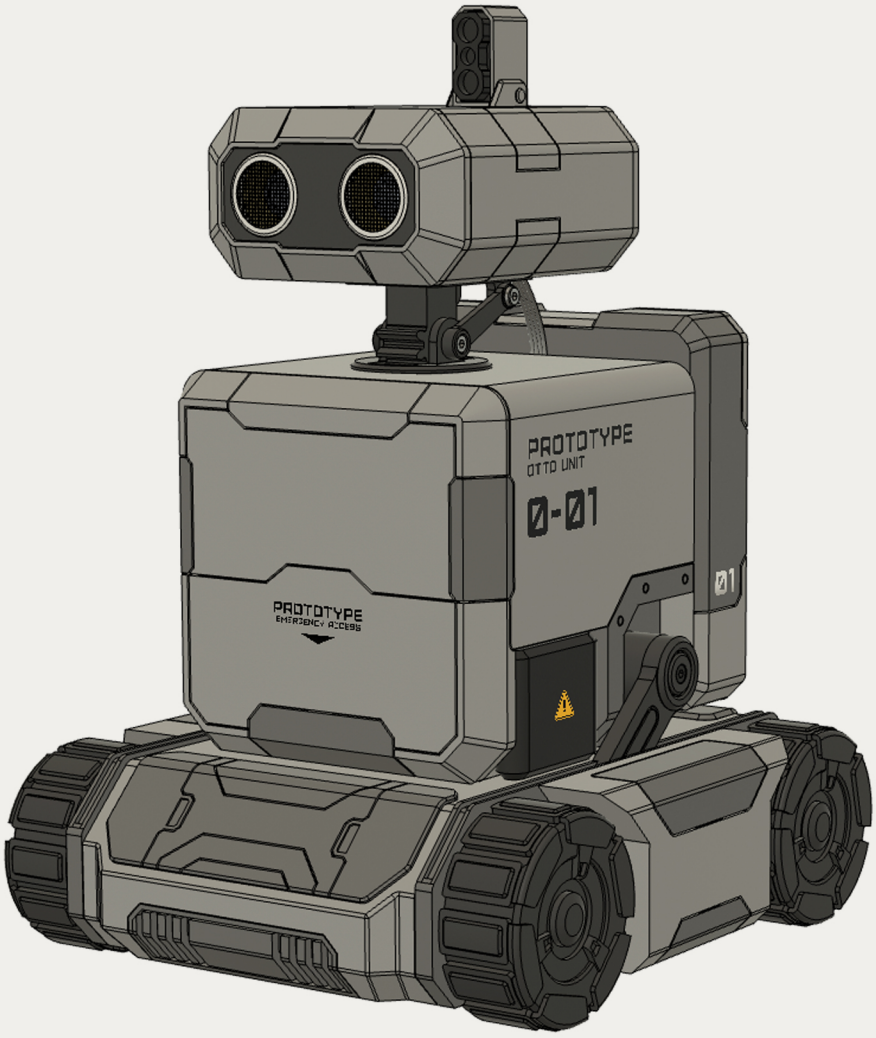


4 PUT IT TOGETHER

Connect a cable to the ultrasonic sensor and snake through the head.
Slide the front in the middle part and click into place.
Finally, slide in the back part to complete the head.

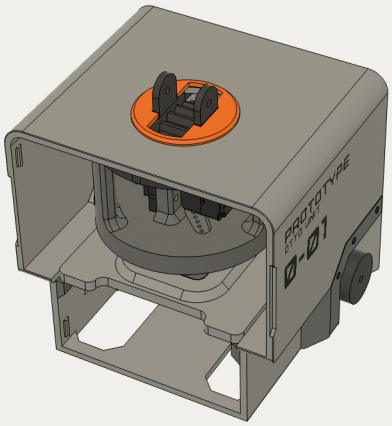


PART 5: FINAL ASSEMBLY



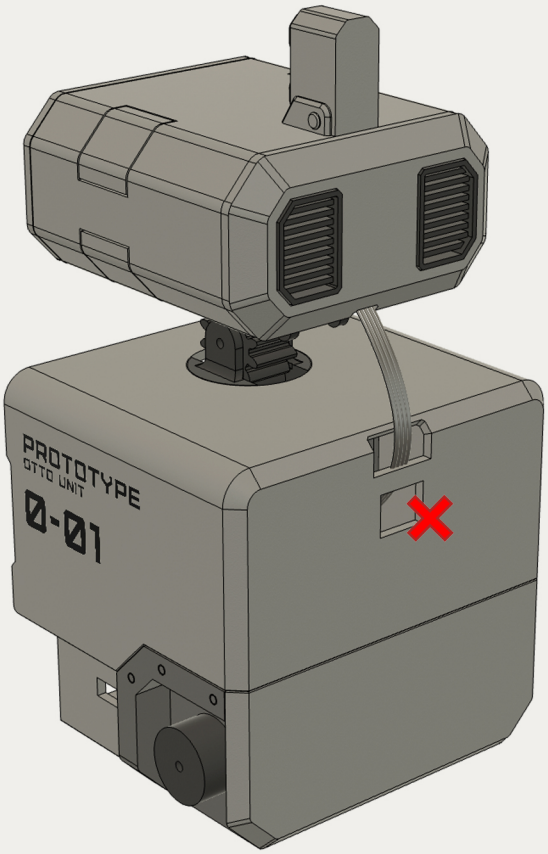
1 HEAD JOINT COVER

Put the cover on the head joint.



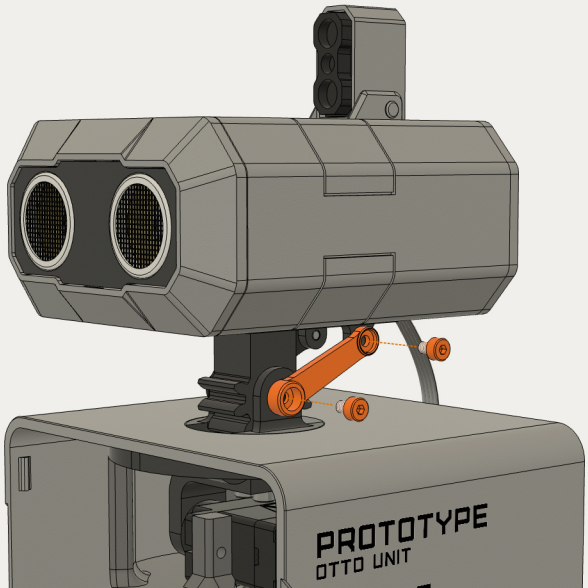
2 CABLE ROUTING

Place the head on the head joint and route the ultrasonic cable through the TOP hole on the back of main body.



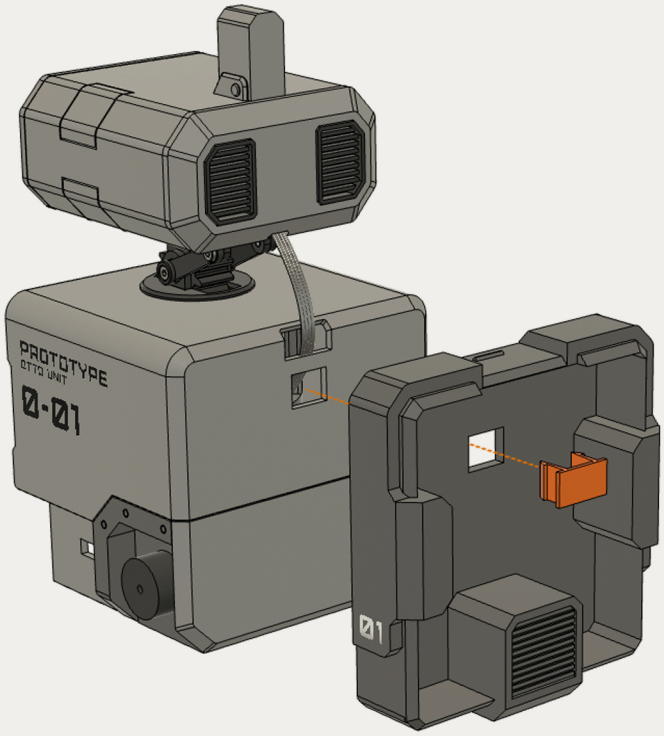
3 SCREW IN HEAD

At last, place the head joint arms on both sides and screw in with M2 screws.

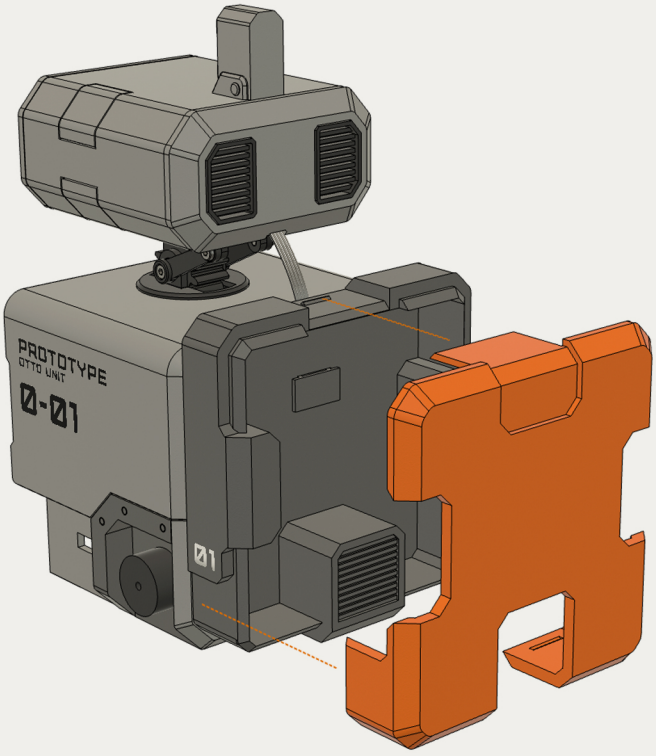


4 CONNECT BACKPACK

Using the small clip, connect the backpack to the body using the BOTTOM hole.



Finally, slide on the backpack cover.

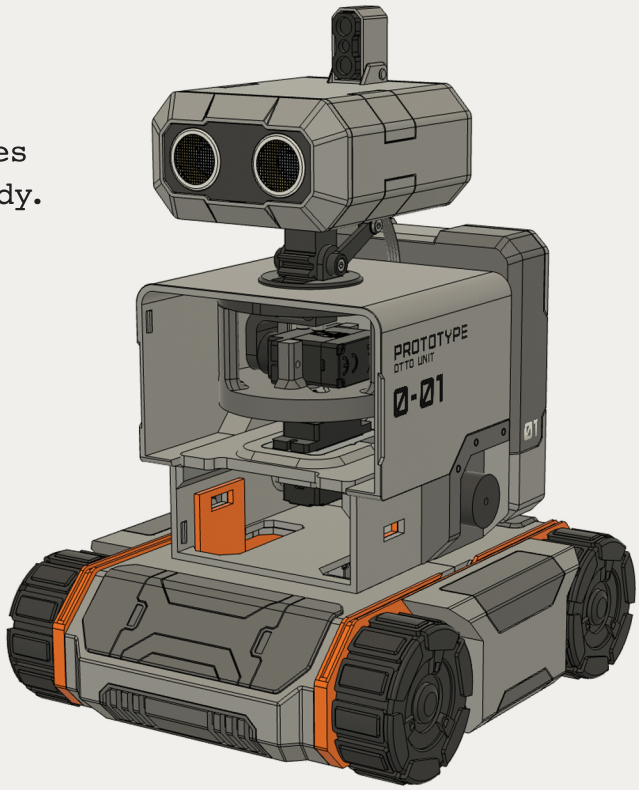


5 MOUNT BODY

Route the ultrasonic and servo cables throught the bottom holes of the body.

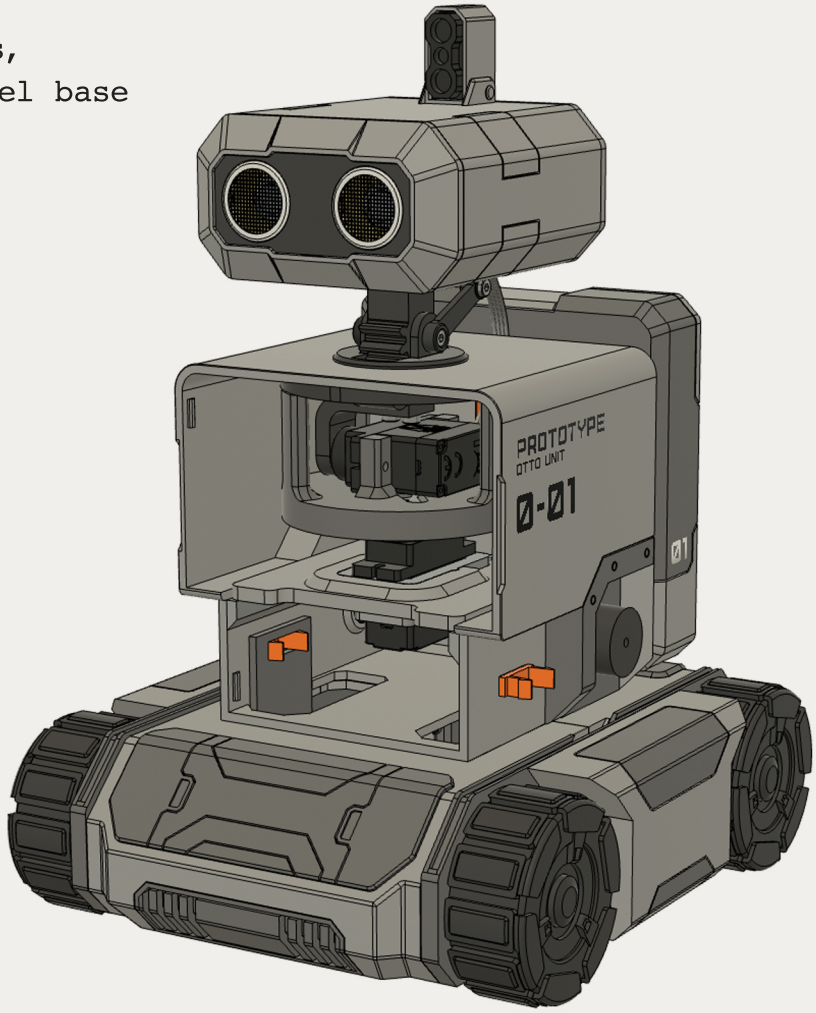
Connect PAN to connector 8.
Connect TILT to connector 9.
Connect ULTRASONIC to connector 1.

Slide the body on the wheel base so that the mounting plates slide through and the holes align.



6 CONNECT BODY

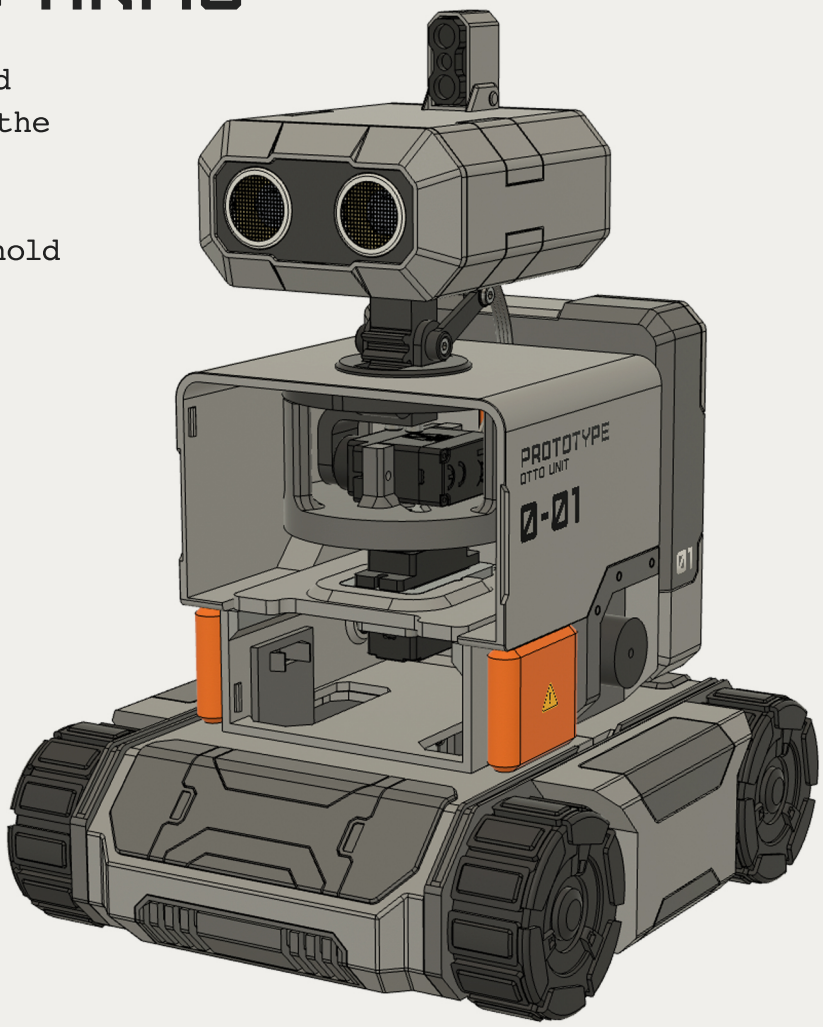
Using the double sides clips, connect the body to the wheel base using the LONGER PRONGS.

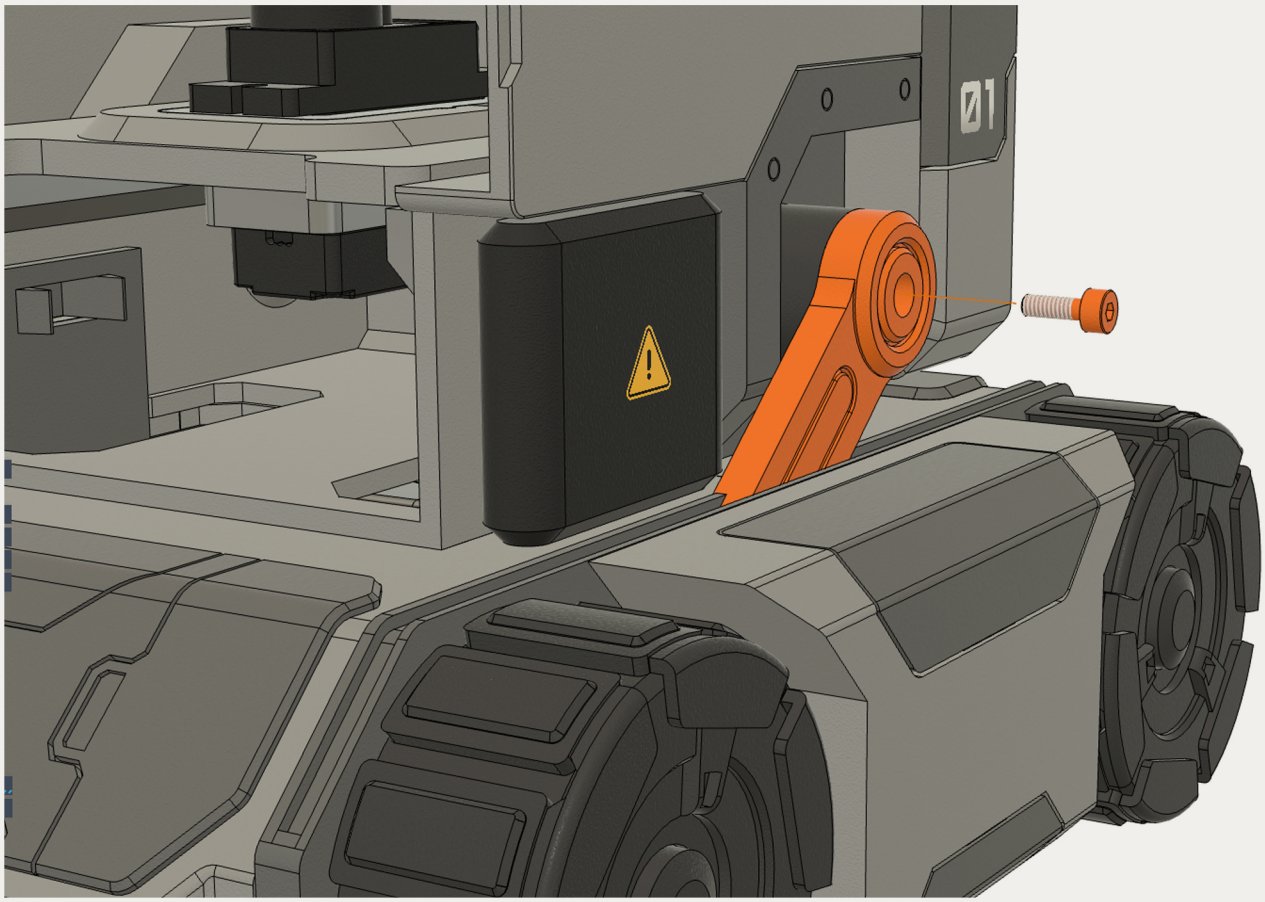


7 TANKS AND ARMS

Using the previously inserted double sided clips, clip in the tanks on both sides.

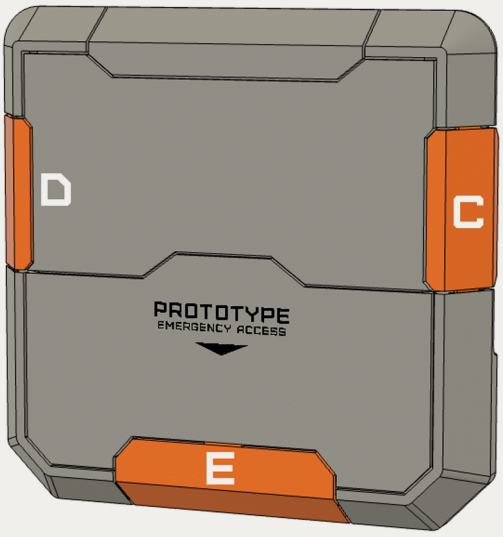
Then slide in the arms and hold in place with M2 screws.





8 FRONT COVER

Assemble the front cover using panels C, D and E.



9 FINAL STEP

Slide in the assembled front panel into the body until it clicks into place.

And now your OTTO UNIT 01 is complete!